
TCX2005: Information Systems, Management and Organisations

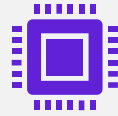
L2: Information Systems in Global Business Today-II

Prakash C. Sukhwal
Senior Lecturer, DISA
pcsdisa@nus.edu.sg

Lesson Objectives



What are Business Model, Business Process, and IS (information systems)?



Why is IS essential for business?



How do IS serve the different management groups in a business?



How do IS that link the enterprise improve organization performance?

Learning Resources



- Textbook (read before class)
 - Read chapter 1 and 2
- Any Other Material
 - Shared on Canvas for this week
 - Videos:
 - [Andrew Ng: How AI Systems Could Empower Any Business](#)
 - [US Foodservice Grows Market with Oracle CRM On Demand](#)

Case Study

Read the case study from textbook on -[Sharp Organization](#)

Identify:

- a. The problem
- b. The solution
- c. Outcomes
- d. Key learning point(s)

Note: be ready to share your thoughts

Case Study

Problem:

- Traditional top-down organizational culture with one-way communication
- Improve information flow between senior management and rank-and-file employees
- Limited collaboration and information sharing across the organization
 - Younger employees seeking platforms for open idea sharing
- Financial pressure requiring more effective and profitable operations

Solution:

- Microsoft Yammer: an enterprise social network
- Voluntary adoption approach (flexibility for both professional and personal content)
- Integration with existing Microsoft tools (E.g., SharePoint, Office 365)
- Phased rollout: starting with pilot program → expanding to domestic employees → then international offices

Case Study

Outcome:

- Rapid adoption to over 10,000 users
- Improved two-way communication between management and employees
- Improved cross-departmental communication and knowledge sharing
 - Department leaders use it to collect ideas from staff about latest technology use
 - Smartphone developers can share opinions about functions and user-friendliness throughout the company
- Enhanced ideas and feedback collection for product development

Learning:

- Strategic opportunity for IS deployment to solve organizational communication challenges
 - IS can transform traditional hierarchical structures into more collaborative, networked organizations
 - IS (Microsoft Yammer) can improve organizational operations through better communication and knowledge sharing across hierarchies
 - IS implementation can lead to cultural transformation in traditional organizations (via new communication patterns)
 - Highlights the importance of appropriate implementation strategies (voluntary adoption, flexible usage) in IT success
-

Business Model, Business Process, and IS

1. Understanding Business Models

- Framework for a company to produce, deliver, and sell a product/service to generate wealth
- The "what" (business activities) and "how" of value creation

How to effectively structure and execute a business?

2. Understanding Business Processes

- Detailed, step-by-step activities that execute the business model.
- Operational mechanisms that bring the business model to life.
- The "how" of value delivery



Business Model
Defines vision and strategy



Business Processes
Outlines operational methods

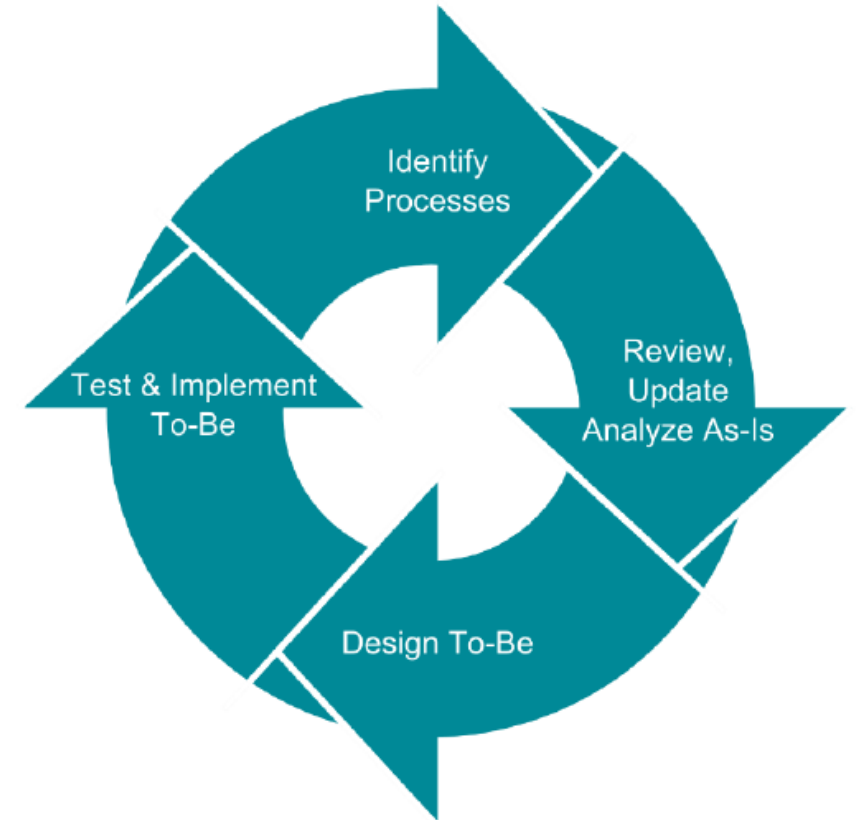
Business Model, Business Process, and IS

- Businesses: Can be seen as collection of business processes
- Business processes may be assets or liabilities
 - E.g., customer credit card approval – “customer experience centered”
 - 3 weeks → 4 days → immediate approval
 - Digital card

Business Model, Business Process and IS

Business process reengineering

- The analysis and design of workflows and business processes within an organization

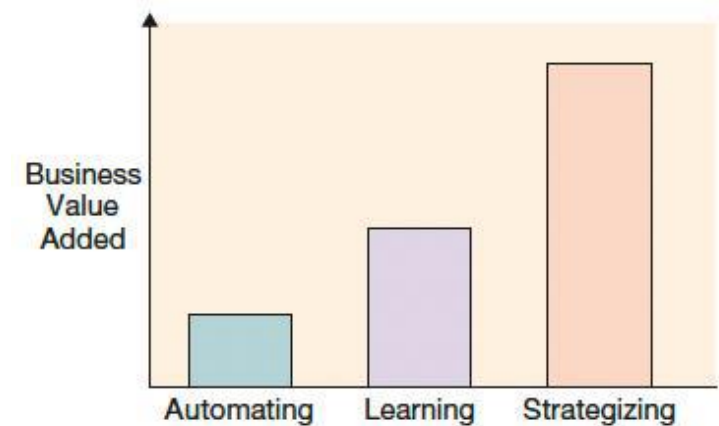


Business Process Reengineering Cycle

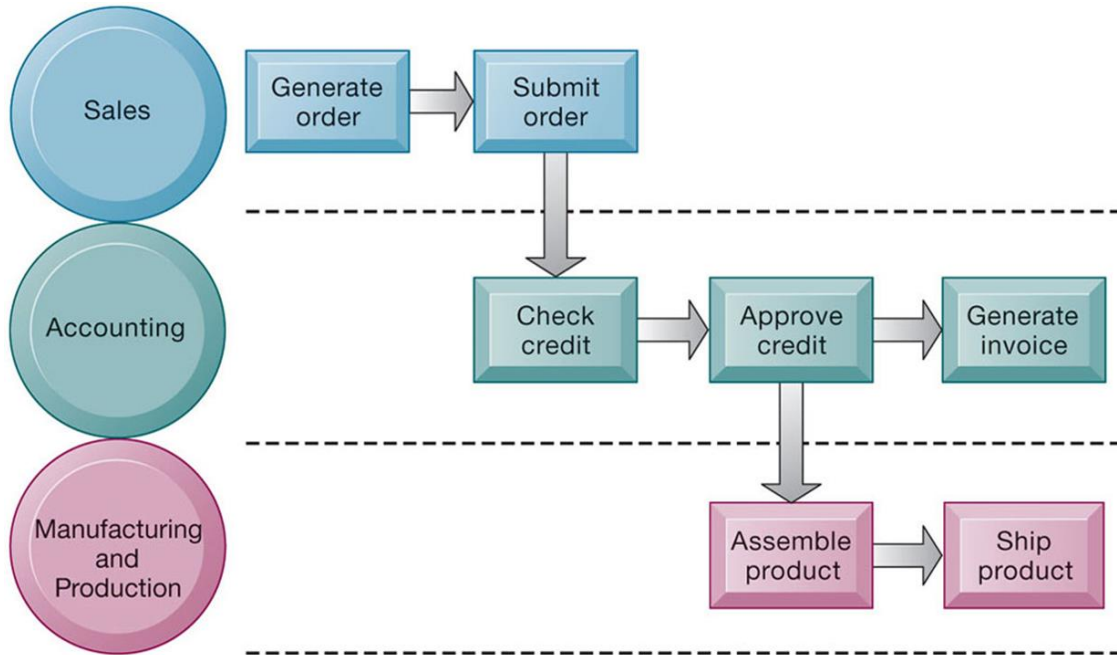
Business Model, Business Process and IS

3. IS as enabler

- Transforms **HOW** companies produce (digital production)
- Changes **HOW** they deliver (digital distribution)
- Revolutionizes **HOW** they sell (online platforms)
- Enhances business processes by:
 - **Automating**: doing things faster
 - **Learning**: doing things better (diffuse, refine, reuse, or innovate knowledge)
 - **Strategizing**: doing things smarter



Business Model, Business Process and IS



- IS improves a process' efficiency
 - Automates manual steps
 - Enables information flow across systems (sequential to parallel)
 - Instantly, from one decision maker to other (avoid delays)
 - Supports decision making and business model

Business Model, Business Process and IS

QUIZ

- Identify as Business Model or Business Process or IS application
 - a. A subscription-based streaming service offered by Netflix.
 - b. ABC Bank uses a centralized database → store customer records, process loan applications → provide online banking services
 - c. Netflix does xyz to manage content acquisition or subscriber billing

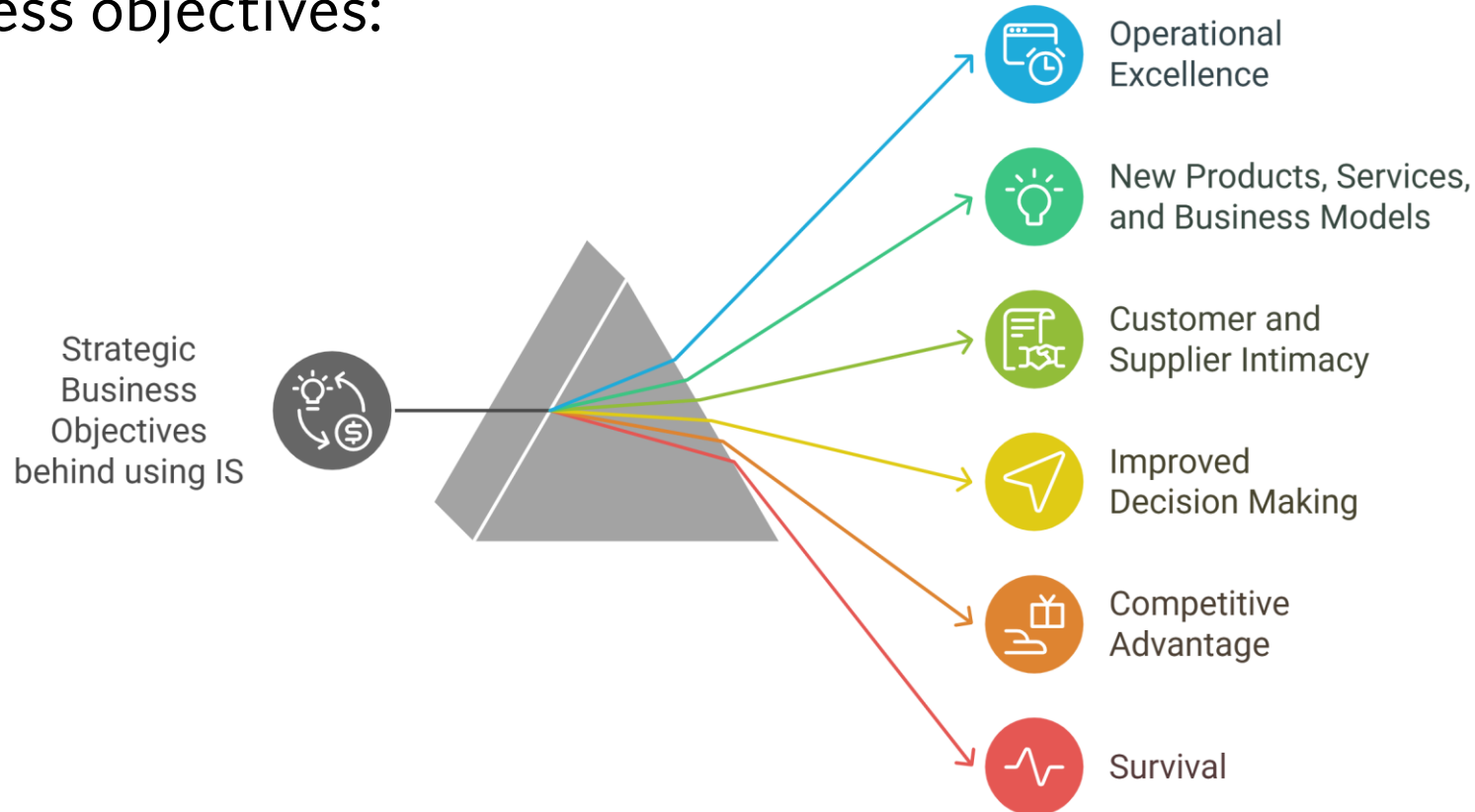
Business Model, Business Process and IS

Quiz-Answers

- a. Business model (This describes the overall strategy for generating revenue and delivering value.)
- b. IS (This involves technology (hardware, software, and data) used to support business operations and decision-making.)
- c. Business process (This is a step-by-step sequence of activities for managing inventory, focusing on operational tasks)

The need for Information Systems

- Firms invest heavily in information systems to achieve six strategic business objectives:



The need for Information Systems

1. Operational Excellence:

- Improve efficiency for higher profitability
- Achieve greater productivity through IS

Example: Walmart's Retail Link System

- Supplier-store connection
- Superior replenishment
- Smart warehouse management



➤ Retail Link allows his suppliers...

Retrieve the sales data of my items in Wal-Mart
Retrieve the inventory data of my items in Wal-Mart
Retrieve Wal-Mart information on its development
Download purchase order on Retail Link

The image shows a login form for the Walmart Retail Link system. It has a dark blue header with the Walmart logo and "Retail Link" text. Below the header, there is a "Log In" section with two input fields: "User ID" and "Password". Below these fields is a "LOG IN" button. Underneath the button is a link that says "Forgot your User ID or Password?". At the bottom of the form is a "CREATE ACCOUNT" button.

[Privacy Policy](#)

The need for Information Systems

Discussion:

- Does Walmart Retail Link create a win-win situation for merchant and suppliers?



➤ Retail Link allows his suppliers...

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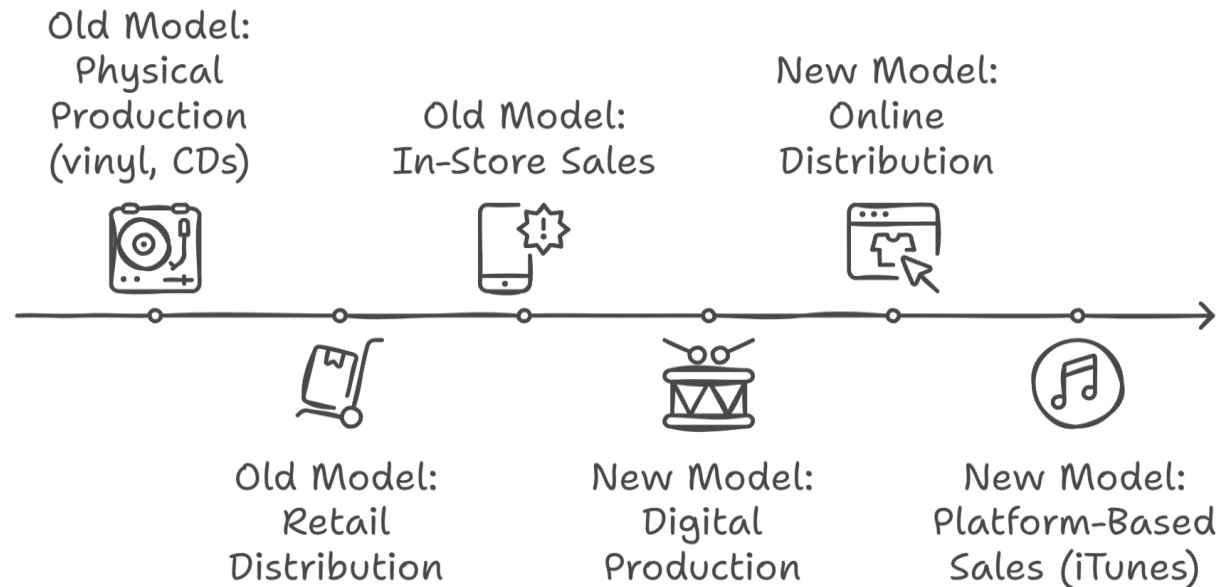
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The need for Information Systems

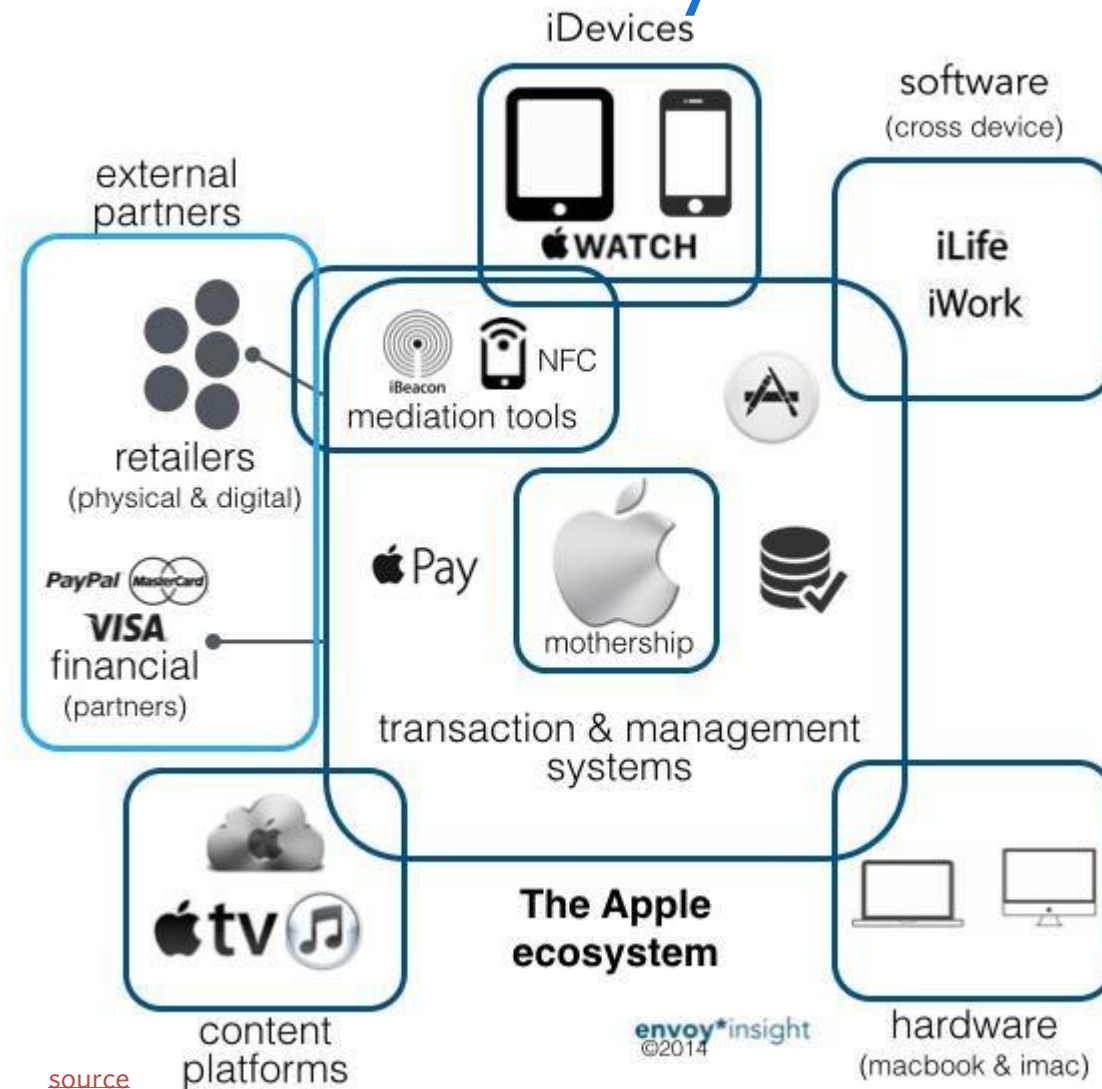
2. New products, services, and business models:

- Create new wealth through IS-enabled business model

Example: Apple Inc. and music distribution



The need for Information Systems

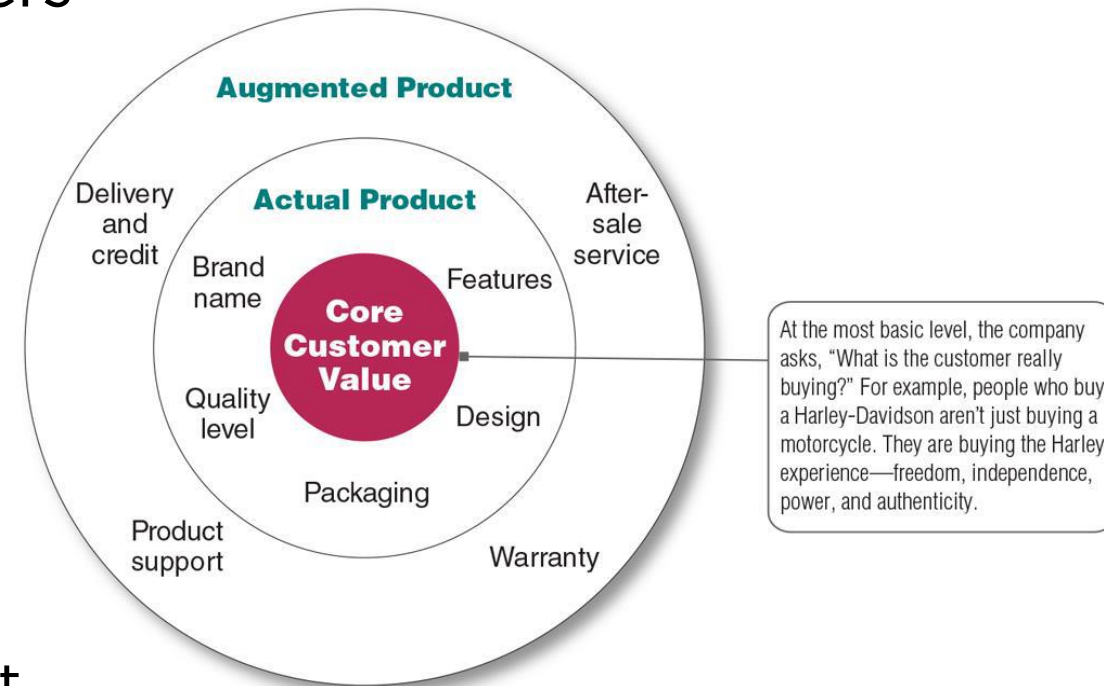


source

The need for Information Systems

3. Customer & Supplier Intimacy

- Serving customer well → Customers returning → Raises revenues and profits
 - Response speed
 - Customer knowledge (Mandarin Oriental Hotels)
- Intimacy with suppliers allows suppliers to provide vital inputs, which lowers costs
 - Real-time inventory management (Charles Tyrwhitt)



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The need for Information Systems

4. Improved Decision Making

- Real-time and accurate information
- Better forecasts & Better planning
- Faster response time

5. Competitive advantage

- Better products and services
- Price competitive
 - Charging less for superior products
- Responding to customers and suppliers in real time

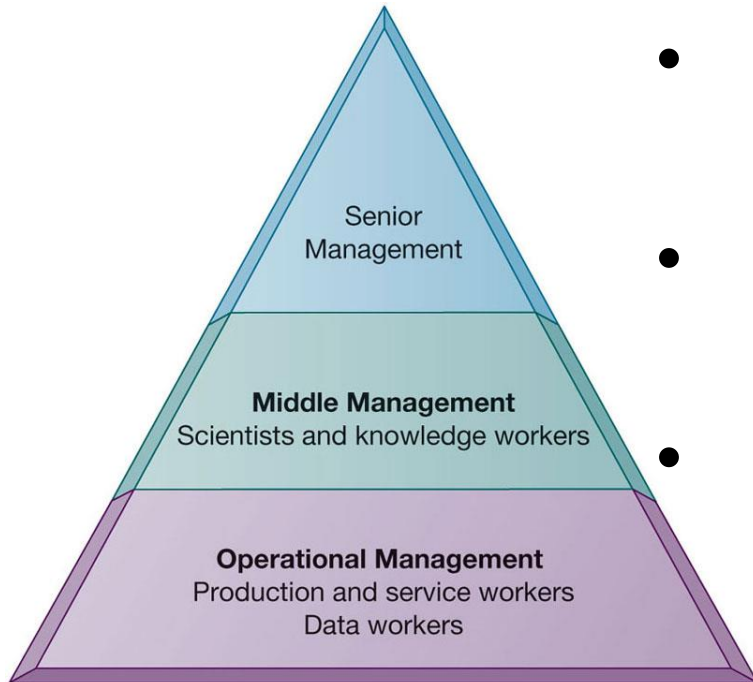
The need for Information Systems

6. Survival

- Necessity for business
 - Changes in industry standards
 - Keep pace with competitors (Barclays ATM case)
 - Societal-level changes (Telecommuting systems/tools)

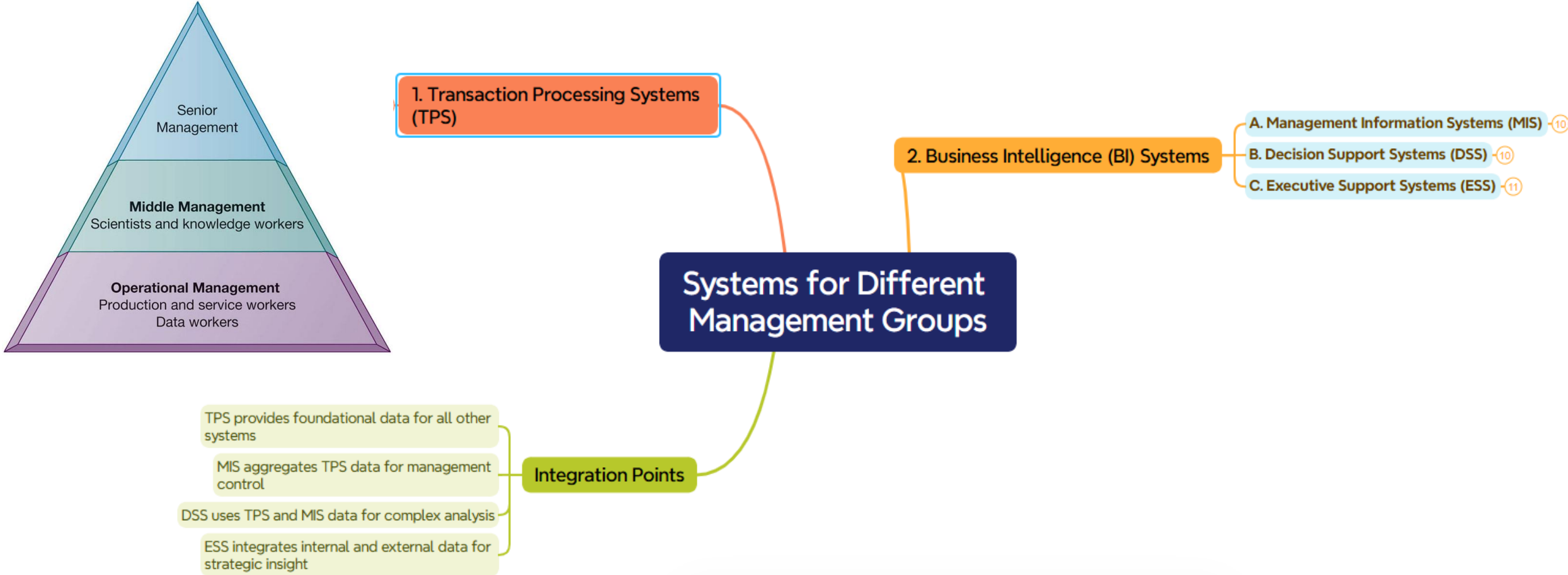
Systems for Different Management Groups

Why?



- Different interests, specialties, and levels require different systems
- No single system can provide all information needs
- Major business functions need dedicated system support:
 - Sales and marketing
 - Manufacturing and production
 - Finance and accounting
 - Human resources

Systems for Different Management Groups



Note: detailed view of the chart is given later in the slides

Systems for Different Management Groups

1. Transaction Processing Systems (TPS)

- Serve operational managers and staff
- Perform and record daily routine transactions necessary to
- Conduct business
 - Examples: sales order entry, payroll, shipping
- Serve predefined, structured goals and decision making

Common features:

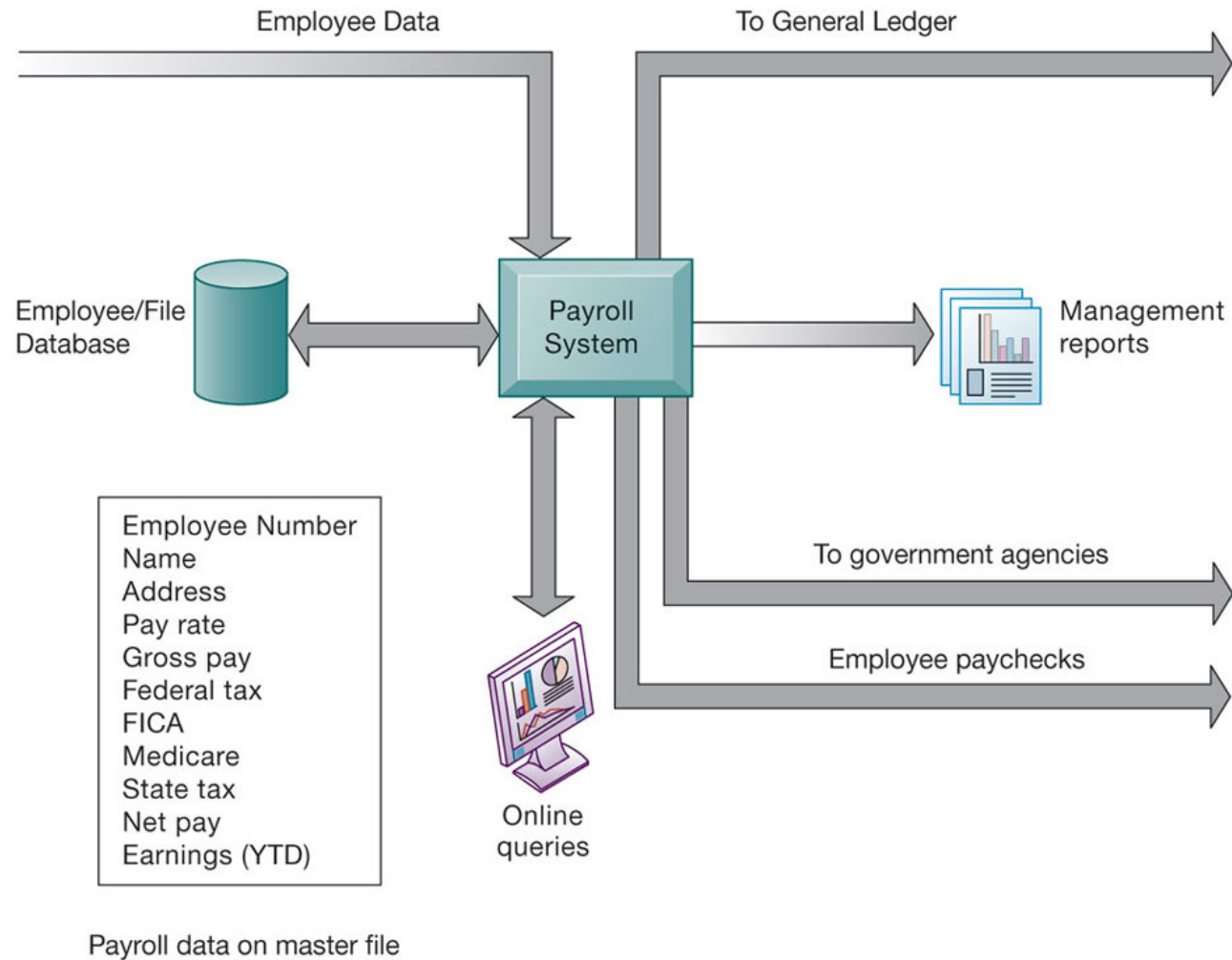
**Simple &
Repetitive**

Systems for Different Management Groups

1. Transaction Processing Systems (TPS)

- Allow managers to monitor status of internal operations and relations with external environment
- Major producers of information for the other systems and business functions
 - Initial data resource for the other systems
 - Importance of accurate and comprehensive data at TPS level
- Central to business

Example



Systems for Different Management Groups

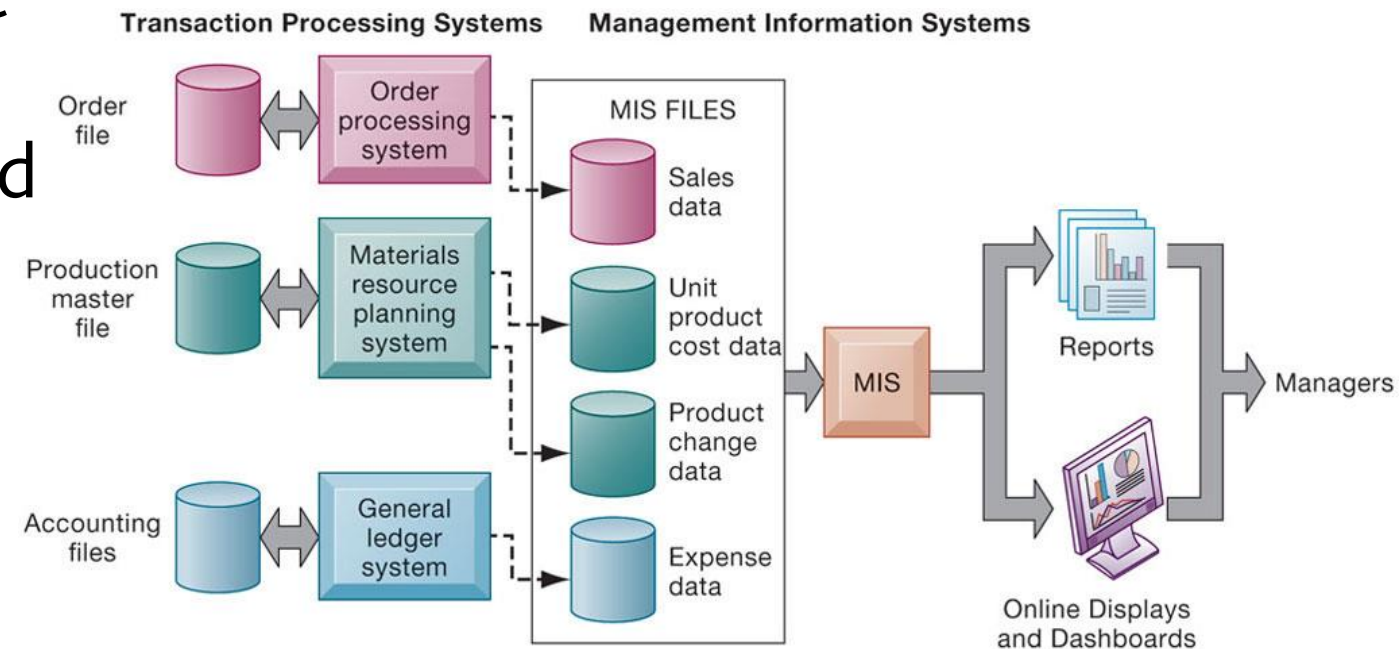
2. Systems for business intelligence (BI)

- Data and software tools to help managers and users make improved decisions
- Support nonroutine decision making
 - E.g., What is the impact on production schedule if sales targets doubled?
- May use external information as well
 - E.g., current stock prices, product prices of competitors
- Sophisticated mathematical models or statistical techniques involved
- Types:
 - A. Management information systems (MIS)
 - B. Decision support systems (DSS)
 - C. Executive support systems (ESS)

Systems for Different Management Groups

A) Management information systems (MIS)

- Serve middle management
- Provide reports on firm's current performance, based on data from TPS
- Provide answers to questions with predefined procedure for answering them
- Typically have little analytic capability



Example MIS Report

E.g., sample report on inpatient admissions for Q3 2024

E.g., Key Performance Indicators:

- **Overall Admission Rate:** 101% of planned
- **Highest Performing Region:** South (105% of target)
- **Most Active Service Line:** General Medicine (56,046 admissions)
- **Service Growth:** General Surgery showing 8% above target in South Region

Service Code	Service Description	Hospital Region	Actual Admissions	Planned Admissions	Actual vs Planned
MED101	General Medicine	Northeast	12,456	13,000	0.96
		South	15,789	15,000	1.05
		Midwest	14,234	14,500	0.98
		West	13,567	13,000	1.04
		Total	56,046	55,500	1.01
SUR202	General Surgery	Northeast	8,345	8,500	0.98
		South	10,234	9,500	1.08
		Midwest	9,123	9,000	1.01
		West	8,890	9,000	0.99
		Total	36,592	36,000	1.02
CAR303	Cardiology	Northeast	5,678	6,000	0.95
		South	7,234	7,000	1.03
		Midwest	6,789	6,500	1.04
		West	5,987	6,000	1.00
		Total	25,688	25,500	1.01

Example MIS Report

What is the use of such a report?

- Contains internal organizational data (can also have external data such as industry benchmarks)
- Focuses on structured, periodic reporting
- Provides summarized data aggregated from operational data
- Shows performance against planned targets
- Can be used by middle management for monitoring and control
- Supports tactical decision-making

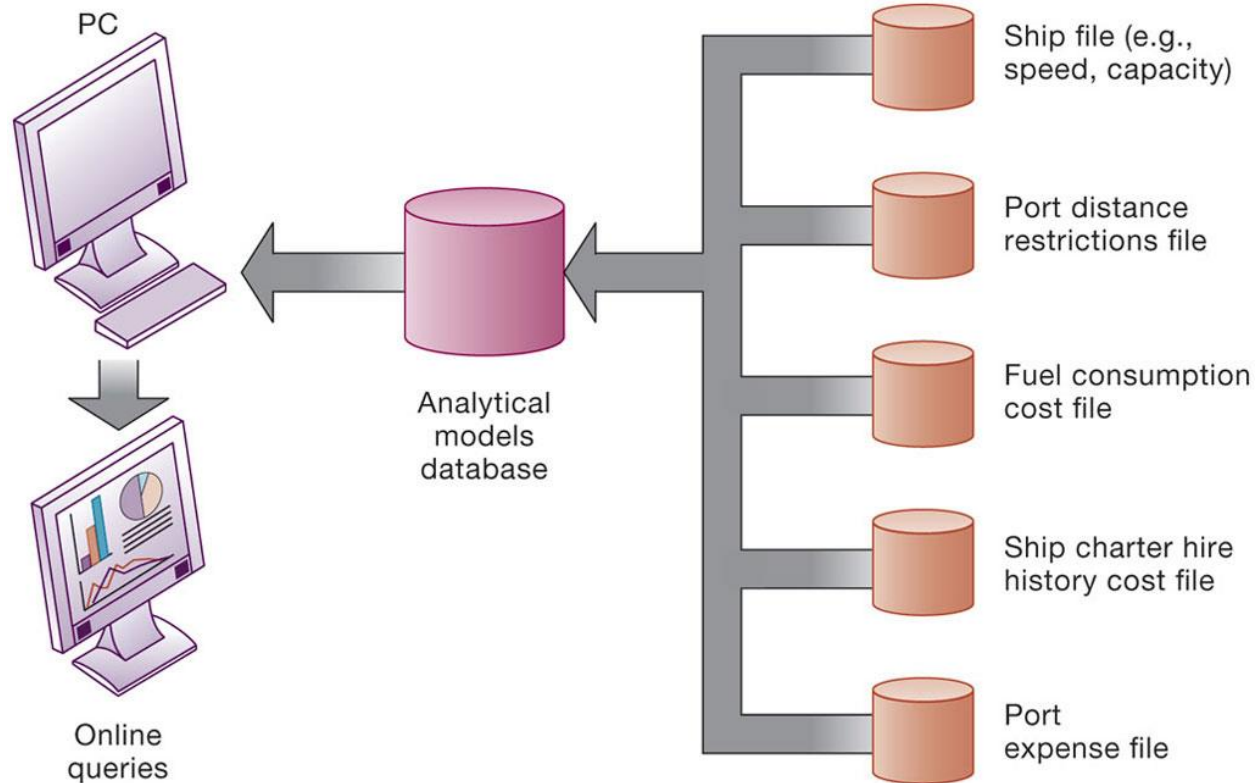
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Systems for Different Management Groups

B) Decision Support Systems (DSS)

- Serve middle management
- Support nonroutine decision making
 - Example: What is the impact on production schedule if December sales doubled?
- May use external information as well TPS / MIS data
- Model driven DSS: use statistical or simulation models
 - Voyage-estimating systems
- Data driven DSS: analyze large volumes of structured data from various sources
 - Marketing analysis systems
 - Analyzes customer behavior patterns by integrating multiple data sources. Such as customer demographics, purchase history, resort usage patterns, and other seasonal trends

Example DSS



Voyage-estimating Systems

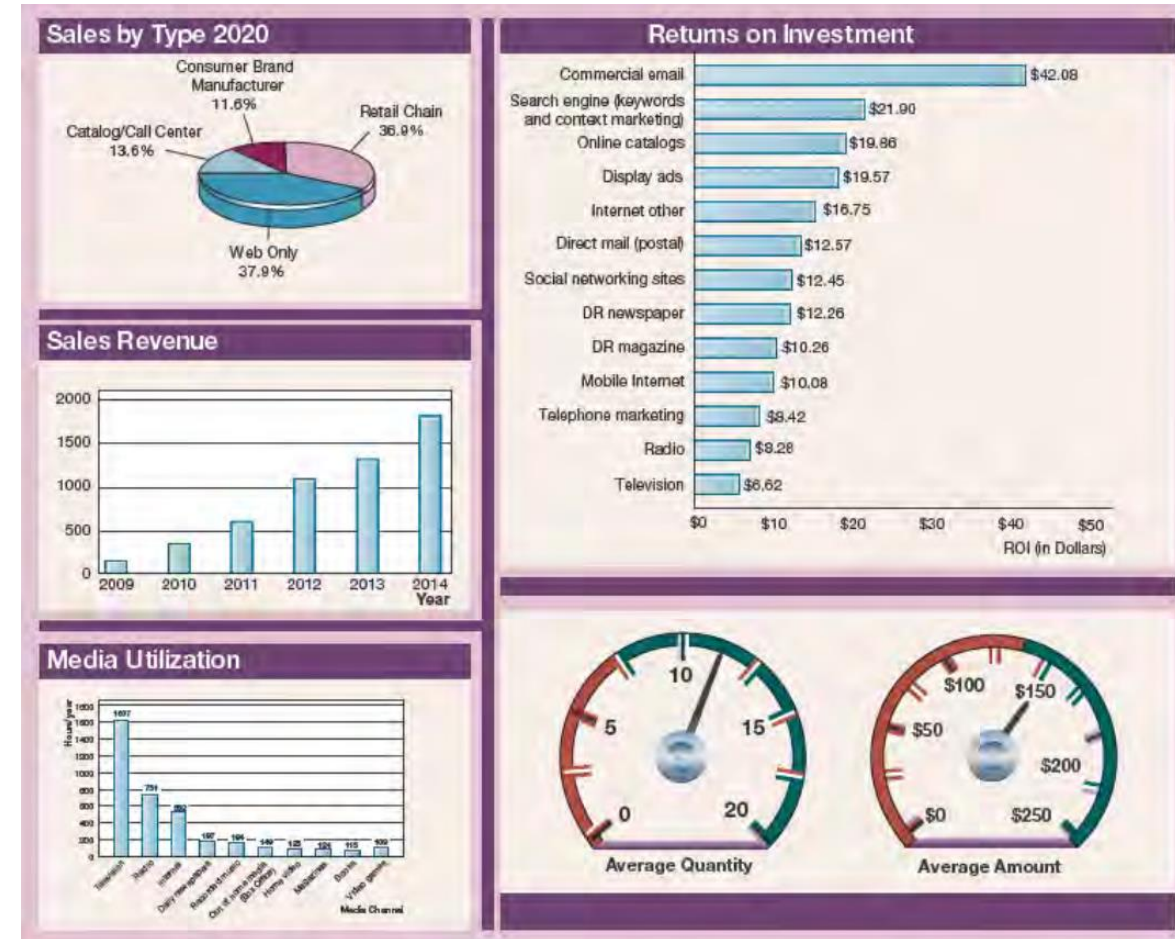
Used by shipping companies

- Calculates costs, risks, and timing
- Variables include:
 - Fuel consumption
 - Port fees
 - Weather conditions
 - Alternative routes

Systems for Different Management Groups

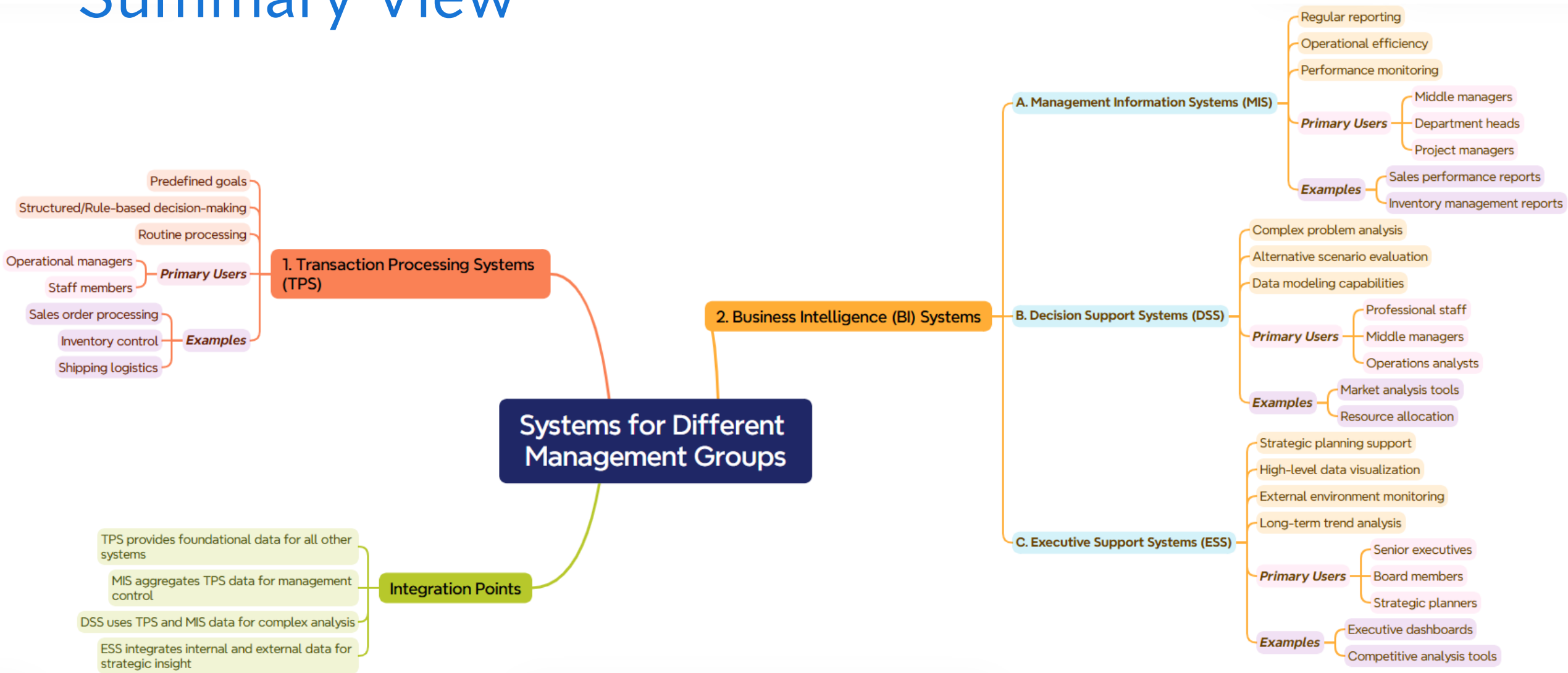
C) Executive Support Systems (ESS)

- Support **senior** management
- Address **nonroutine** decisions
 - Requiring judgment, evaluation, and insight
- Incorporate data about **external events** (e.g., new tax laws or competitors) as well as summarized information from **internal MIS and DSS**
 - Example: **Digital dashboard** with real-time view of firm's financial performance



A digital dashboard displays graphs and charts of key performance indicators for managing a company.

Summary View



Quiz

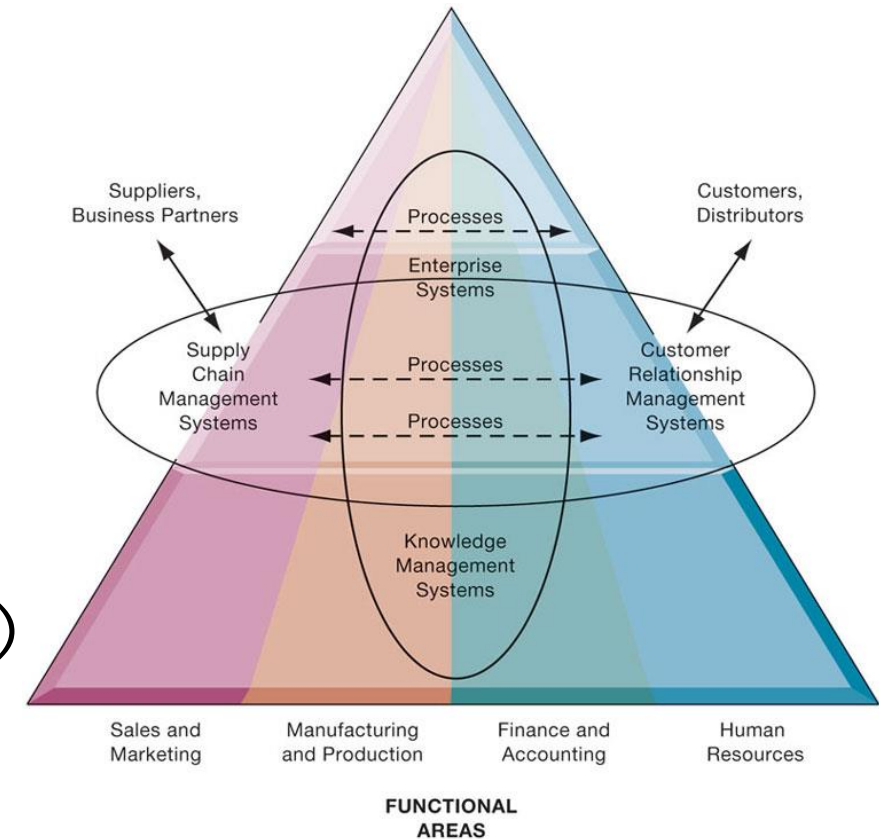
A manufacturing company uses a system that allows managers to ask questions like "What would happen to our production capacity if we added another production shift?" and "How would raw material costs affect our pricing?"

This is an example of:

- MIS
- TPS
- DSS
- ESS

Enterprise Applications

- Systems for linking the enterprise
- Span functional areas
- Execute business processes across firm
- Include all levels of management
- Four major applications:
 - Enterprise systems (**ERP**)
 - Supply chain management systems (**SCM**)
 - Customer relationship management systems (**CRM**)
 - Knowledge management systems (**KMS**)

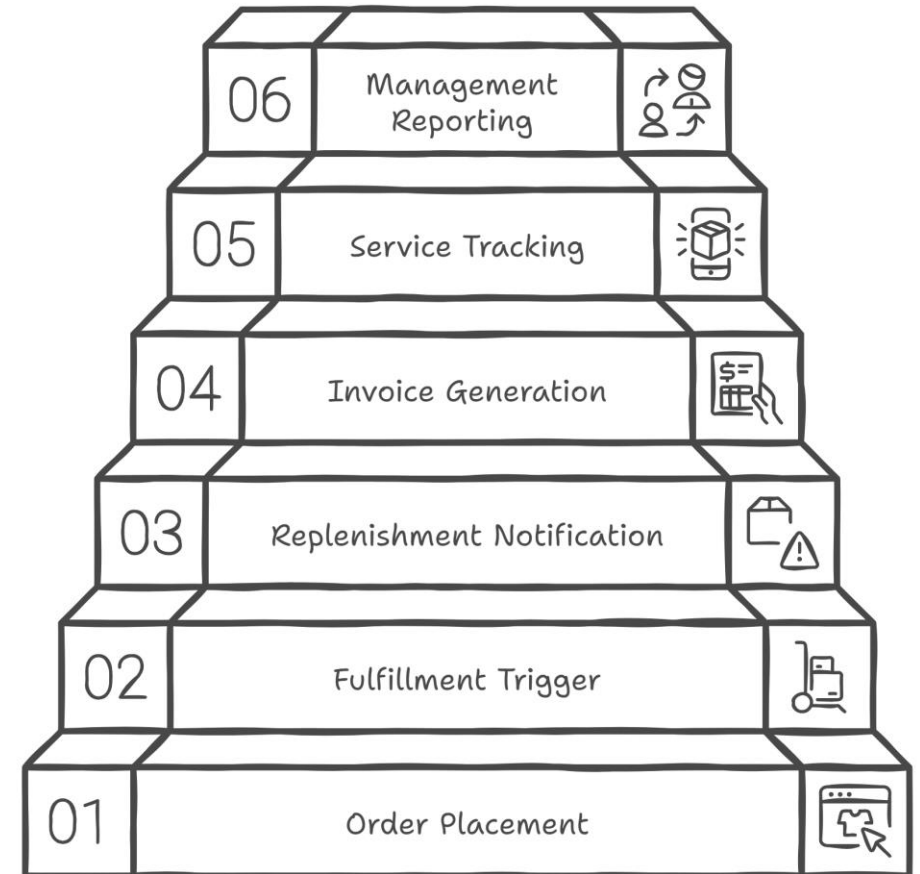


Enterprise Applications

Enterprise systems (ERP)

- Integrate all business processes into a single unified software system (manufacturing and production, finance and accounting, sales and marketing, human resources)
- Create comprehensive data repository for fragmented business operations

Example ERP
Process Flow

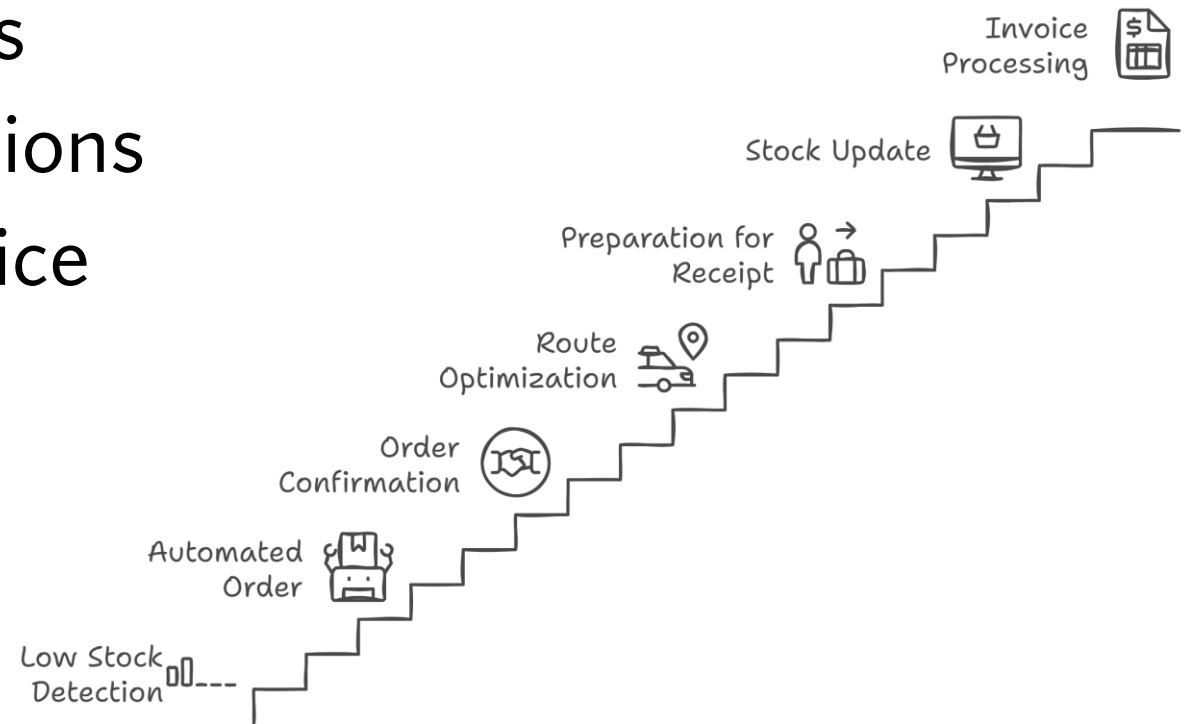


Enterprise Applications

Supply Chain Management Systems (SCM)

- Manage supplier relationships
- Optimize supply chain operations
- Enable efficient product/service delivery
- Benefits
 - Reduced operational costs
 - Optimized delivery times
 - Improved sourcing decisions
 - Enhanced production scheduling

Example
SCM Process Flow



Enterprise Applications

Homework

- Self-Read about the other 2 applications (CRM, KMS) from the recommended textbook